Fall 2025, EE 582: Assignment 2:
Nodal-Analysis-based Modeling of Electric Circuits
Due on Tuesday, October 7 (submission in Canvas)

REPORTING

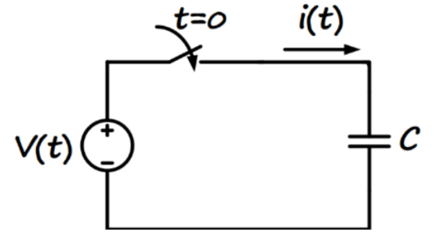
<u>All reports should be typed.</u> The assignment report should demonstrate the individual's work and include printouts of the model windows, simulation results, as well as answers, discussions, and conclusions wherever appropriate. Use graphical commands such as **`hold`** and **`subplot`** for compact and clear presentation of the figures in your report. The discussions and comments should not be long (**avoid writing unnecessarily long reports!**), but they should demonstrate your understanding of the problem.

<u>Please provide your MATLAB codes as part of your report.</u>

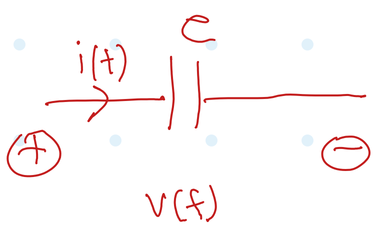
Best of luck!

SECTION A (Numerical Oscillations of Discretization Rules)

One way to analyze the properties of various discretization methods (e.g., in terms of their numerical oscillations) is to solve a lossless first-order system (e.g., inductance or capacitance) when the state variable of the component is suddenly changed. The circuit on the right considers a case where a dc voltage is suddenly applied to a capacitor. Because the source voltage is applied like a step function, we know that the current of the capacitor should look like an impulse, analytically. Let us assume that in the simulations all the variables are zero at t=0 and the voltage source takes its value at t=Δt.



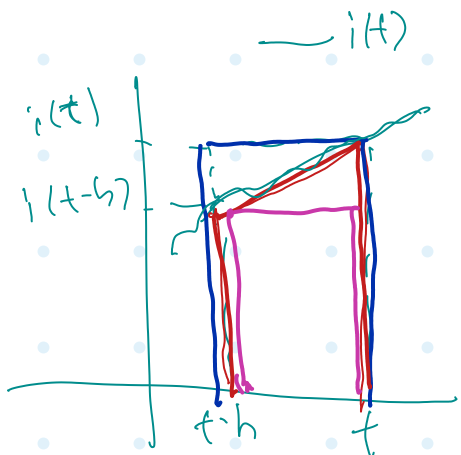
1) Determine the step-by-step solution for the current in the capacitor (in terms of capacitance C, time-step size Δt, and source voltage) using the following discretization rules: a) Forward Euler, b) Backward Euler, c) Trapezoidal.



$$C\frac{dV(t)}{dt} = i(t)$$

$$\to \int_{t-h}^{t} dV(t) = \frac{1}{C}\int_{t-h}^{t} i(t)\,dt$$

$$\to \boxed{V(t) = V(t-h) + \frac{1}{C}\int_{t-h}^{t} i(t)\,dt}$$



Area $A = \frac{h}{2}\left(i(t)+i(t-h)\right)$

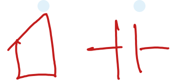
$A = h \cdot i(t-h)$

$A = h \cdot i(t)$

$$v(t) = v(t-h) + \frac{1}{C}\cdot\frac{h}{2}\cdot\left(i(t) + i(t-h)\right)$$

or $$v(t) = v(t-h) + \frac{h}{2C}\, i(t-h) + \frac{h}{2C}\, i(t)$$

or $$v(t) = R_c\, i(t) + e_h(t)$$



$$R_c = \frac{h}{2C} \qquad e_h(t) = R_c\, i(t-h) + v(t-h)$$ (1a)

$$v(t) = v(t-h) + \frac{h}{C}\, i(t)$$

or $$v(t) = R_c\, i(t) + e_h(t)$$

$$R_c = \frac{h}{C} \qquad e_h(t) = v(t-h)$$

$$v(t) = v(t-h) + \frac{h}{C}\, i(t-h)$$

or $$v(t) = e_h(t)$$

$$R_c = \frac{h}{C} \qquad e_h(t) = v(t-h) + R_c\, i(t-h)$$

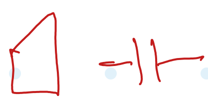
$$i(t) = \frac{1}{R_C}\left(v(t) - i_h(t)\right)$$

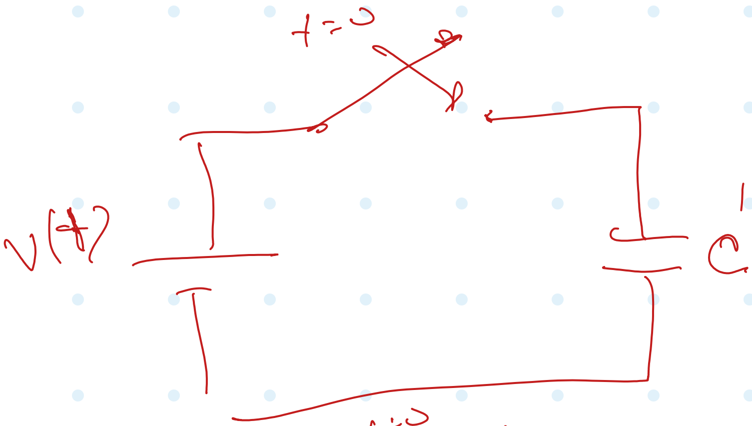
$$i(t) = \frac{1}{R_C}\left(v(t) - v(t-h) - R_C \cdot i(t-h)\right)$$

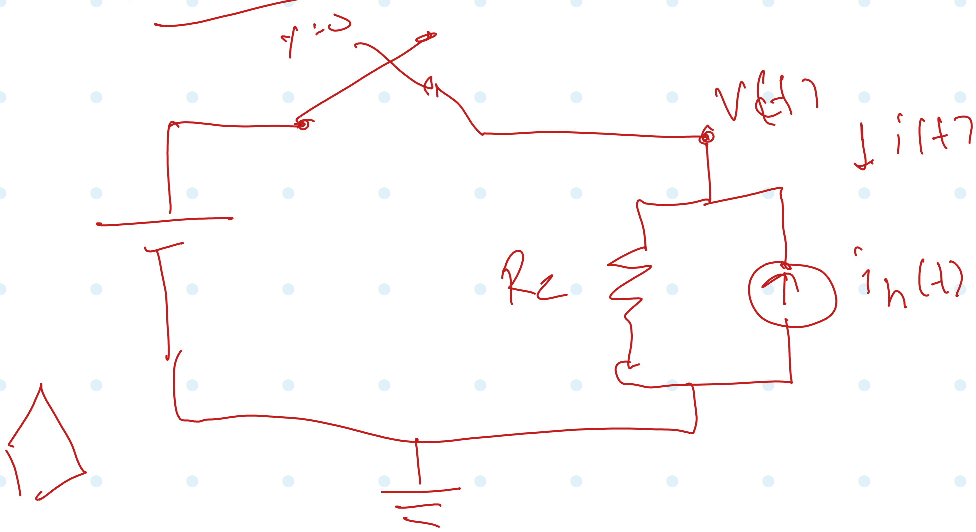
$$i(t) = \frac{1}{R_C} v(t) - i_h(t)$$

$$R_C = \frac{h}{2C} \qquad i_h(t) = \frac{1}{R_C} v(t-h) + i(t-h)$$

1b







* assume

t	$v(t)$	$i(t)$	e_n	i_n
$-h$	V_o	0	—	—
0^+	V_{DC}	$\frac{V_{DC}}{R_C} - \frac{V_o}{R_C}$	V_o'	V_o/R_C
h	V_{DC}	$\frac{V_o}{R_C} - \frac{V_{DC}}{R_C}$	$2V_{DC} - V_o$	$\frac{2V_{DC}}{R_C} - \frac{V_o}{R_C}$
$2h$	V_{DC}	$\frac{V_{DC}}{R_C} - \frac{V_o}{R_C}$	V_o	V_o/R_C
$3h$	V_{DC}	$\frac{V_o}{R_C} - \frac{V_{DC}}{R_C}$	$2V_{DC} - V_o$	$\frac{2V_{DC}}{R_C} - \frac{V_o}{R_C}$
$4h$	V_{DC}	$\frac{V_{DC}}{R_C} - \frac{V_o}{R_C}$	V_o	V_o/R_C

∘ $v(t) = R_c i(t) + e_h(t)$

$R_c = \frac{h}{2C}$ $\quad e_h(t) = R_c i(t-h) + v(t-h)$

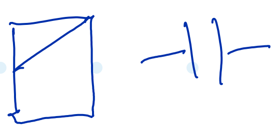
∘ $i(t) = \frac{1}{R_c} v(t) - i_h(t)$

1b

$R_c = \frac{h}{2C}$ $\quad i_h(t) = \frac{1}{R_c} v(t-h) + i(t-h)$

∘ $i_n(t) = \frac{e_n(t)}{R_c}$

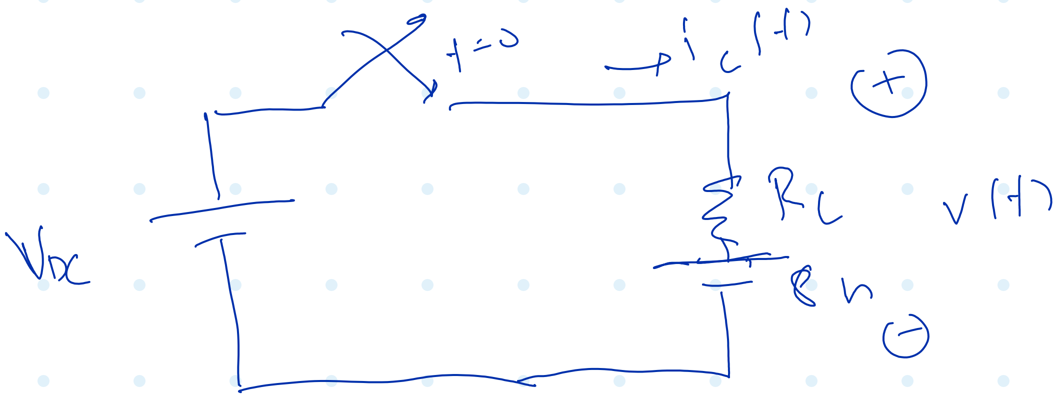
$$v(t) = R_c\, i(t) + e_h(t)$$



$$R_c = \frac{h}{C} \qquad e_h(t) = v(t-h)$$

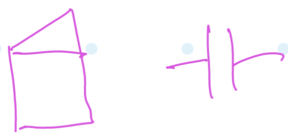
$$i(t) = \frac{1}{R_c} v(t) - i_h(t)$$

$$i_h(t) = \frac{v(t-h)}{R_c} = \frac{e_h(t)}{R_c}$$



t	$v(t)$	$i(t)$	$e_h = v(t-h)$	$i_h = \frac{1}{R_c} v(t-h)$
$-h$	V_0	0	—	—
0^+	V_{DC}	$\frac{V_{DC}}{R_c} - \frac{V_0}{R_c}$	V_0	$\frac{V_0}{R_c}$
h	V_{DC}	0	V_{DC}	$\frac{V_0}{R_C}$
$2h$	V_{DC}	0	V_{DC}	$\frac{V_{DC}}{R_C}$
$3h$	V_{DC}	0	V_{DC}	$\frac{V_{DC}}{R_C}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	
$4h$	V_{DC}	0	V_{DC}	$\frac{V_{DC}}{R_C}$

- $v(t) = \rho_h(t)$



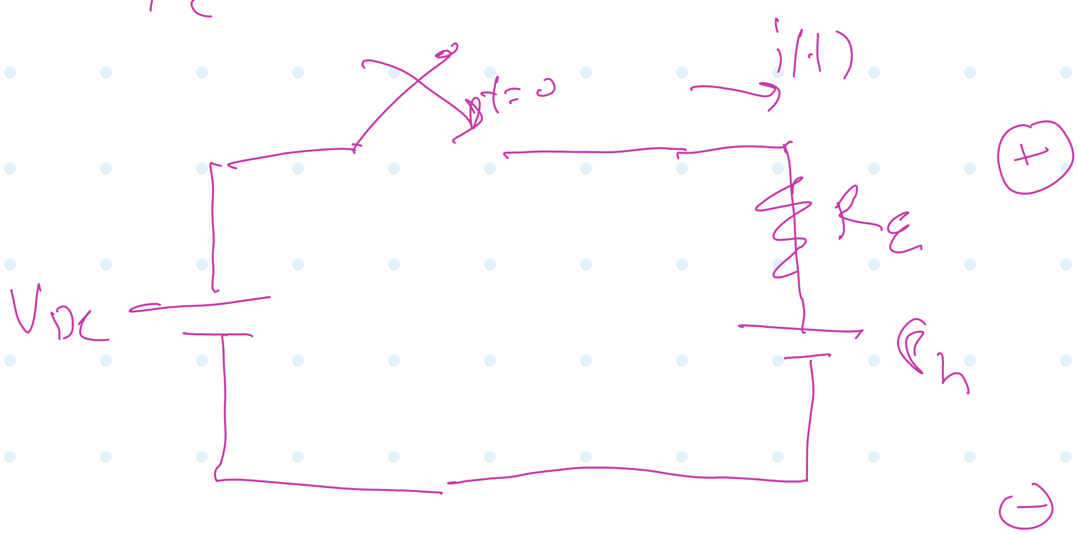
$R_c = \underline{h}$ $\qquad \rho_h(t) = v(t-h) + R_c i(t-h)$

$$i(t) = \frac{v(t) - \rho_h}{R_\varepsilon}$$

$\nearrow +\infty$ if $\rho_h < v(t)$

$\searrow -\infty$ if $\rho_h > v(t)$

$$i_h(t) = \frac{v(t-h)}{R_\varepsilon} + i(t-h) \qquad i(t) = \frac{1}{R_\varepsilon} - i_h(t)$$



$$i(t) = \frac{1}{R_\varepsilon} v(t) - \frac{\rho_h}{R_\varepsilon}$$

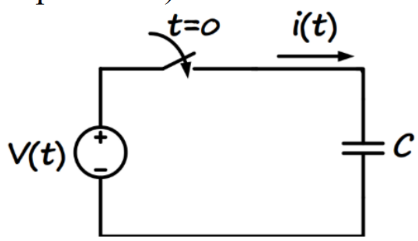
- $i(t) = \frac{1}{R_\varepsilon} v(t) - i_h$

$$i_h = \frac{\rho_h}{R_\varepsilon} = \frac{v(t-h)}{R_\varepsilon} + \frac{R_c i(t-h)}{R_\varepsilon}$$

t	$v(t)$	$i(t) = \dfrac{v(t) - e_h(t)}{R_\varepsilon}$	$e_h(t) = v(t-h) + R_c\, i(t-h)$	$i_h(t) = \dfrac{e_h}{R_\varepsilon}$
$-h$	V_0	0	—	—
0^+	V_{DC}	$\dfrac{V_{DC}}{R_\varepsilon} - \dfrac{V_0}{R_\varepsilon}$	V_0	$\dfrac{V_0}{R_\varepsilon}$
h	V_{DC}	$V_0\left(\dfrac{R_c}{R_\varepsilon^2}\right) - V_{DC}\left(\dfrac{R_c}{R_\varepsilon^2}\right)$	$V_{DC}\left(1 + \dfrac{R_c}{R_\varepsilon}\right) - V_0\left(\dfrac{R_c}{R_\varepsilon}\right)$	$V_{DC}\left(\dfrac{1}{R_\varepsilon} + \dfrac{R_c}{R_\varepsilon^2}\right) - V_0\left(\dfrac{R_c}{R_\varepsilon^2}\right)$
$2h$	V_{DC}	$V_{DC}\left(\dfrac{R_c^2}{R_\varepsilon^3}\right) - V_0\left(\dfrac{R_c^2}{R_\varepsilon^3}\right)$	$V_{DC}\left(1 - \dfrac{R_c^2}{R_\varepsilon^2}\right) + V_0\left(\dfrac{R_c^2}{R_\varepsilon^2}\right)$	$V_{DC}\left(\dfrac{1}{R_\varepsilon} - \dfrac{R_c^2}{R_\varepsilon^3}\right) + V_0\left(\dfrac{R_c^2}{R_\varepsilon^3}\right)$
$3h$	V_{DC}	$V_0\left(\dfrac{R_c^3}{R_\varepsilon^4}\right) - V_{DC}\left(\dfrac{R_c^3}{R_\varepsilon^4}\right)$	$V_{DC}\left(1 + \dfrac{R_c^3}{R_\varepsilon^3}\right) - V_0\left(\dfrac{R_c^3}{R_\varepsilon^3}\right)$	$V_{DC}\left(\dfrac{1}{R_\varepsilon} + \dfrac{R_c^3}{R_\varepsilon^4}\right) - V_0\left(\dfrac{R_c^3}{R_\varepsilon^4}\right)$

SECTION A (Numerical Oscillations of Discretization Rules)

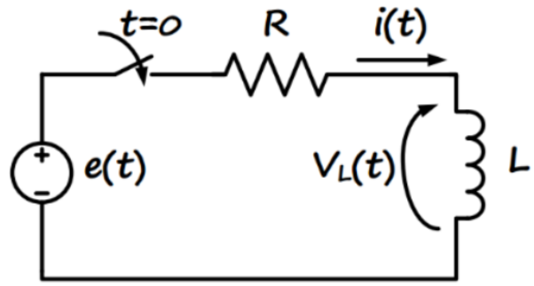
One way to analyze the properties of various discretization methods (e.g., in terms of their numerical oscillations) is to solve a lossless first-order system (e.g., inductance or capacitance) when the state variable of the component is suddenly changed. The circuit on the right considers a case where a dc voltage is suddenly applied to a capacitor. Because the source voltage is applied like a step function, we know that the current of the capacitor should look like an impulse, analytically. Let us assume that in the simulations all the variables are zero at t=0 and the voltage source takes its value at t=Δt.



2) Comment on how the simulation time-step size impacts the numerical solution, and which method is preferred in this case.

SECTION B (Basic Circuit Numerical Simulation)

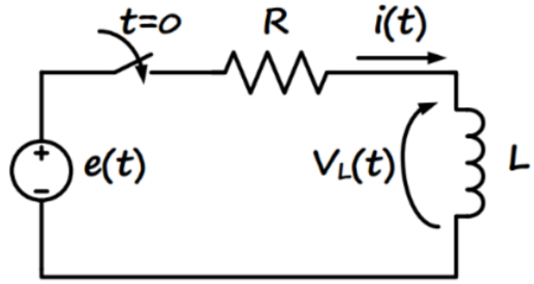
The parameters of the circuit below are: $R = 10\ \Omega$, $L = 20$ mH, $e(t) = 10$ V dc, and the switch is closed at t=0. The circuit below is to be solved from t=0 to t=10 ms using: a) PSCAD software, b) your own MATLAB code based on the nodal analysis method.



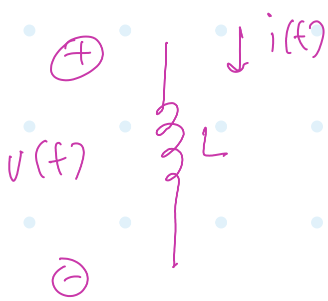
1) Obtain the solution for the inductor current and voltage using PSCAD with time-step sizes Δt=0.1 ms and Δt=0.8 ms.

SECTION B (Basic Circuit Numerical Simulation)

The parameters of the circuit below are: $R = 10\ \Omega$, $L = 20$ mH, $e(t) = 10$ V dc, and the switch is closed at t=0. The circuit below is to be solved from t=0 to t=10 ms using: a) PSCAD software, b) your own MATLAB code based on the nodal analysis method.



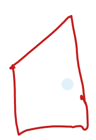
2) Discretize the circuit and write a MATLAB code to solve the circuit step-by-step using the two discretization rules: a) Backward Euler, b) Trapezoidal, each with time-step sizes Δt=0.1 ms and Δt=0.8 ms.



$$L\frac{di(t)}{dt} = v(t)$$

$$\int di(t) = \frac{1}{L}\int v(t)\,dt$$

$$\boxed{i(t) - i(t-h) = \frac{1}{L}\int v(t)\,dt}$$



$$i(t) - i(t-h) = \frac{1}{L}\frac{h}{2}\left(v(t-h) + v(t)\right)$$

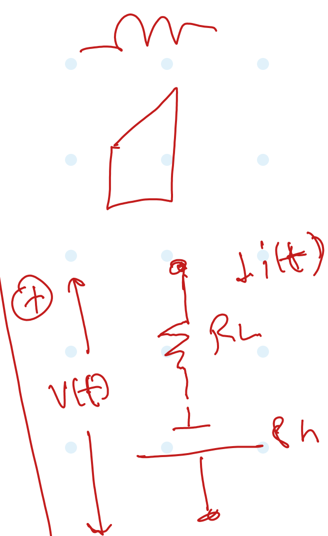
$$\frac{2L}{h}\left(i(t) - i(t-h)\right) - v(t-h) = v(t)$$

$$\boxed{\begin{gathered} v(t) = R_L\, i(t) - e_h \\ R_L = \frac{2L}{h} \qquad e_h = R_L\, i(t-h) + v(t-h) \\ i_h = \frac{e_h}{R_L} = i(t-h) + \frac{1}{R_L} v(t-h) \end{gathered}}$$

$$i(t) = \frac{v(t)}{R_L} + \frac{e_h}{R_L}$$

$$i(t) = \frac{v(t)}{R_L} + i(t-h) + \frac{v(t-h)}{R_L}$$

$$a \quad i(t) = \frac{v(t)}{R_L} + i_h$$



$$v(t) = R_L i(t) - e_h$$

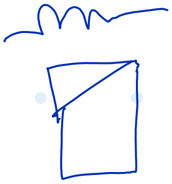
$$i(t) = \frac{v(t)}{R_L} + i_h$$

$$R_L = \frac{2L}{h}$$

$$e_h = R_L i(t-h) + v(t-h)$$

$$i_h = i(t-h) + \frac{v(t-h)}{R_L}$$

$$= e_h / R_L$$



$$i(t) - i(t-h) = \frac{1}{L} \cdot h\, v(t)$$

$$R_L \left(i(t) - i(t-h) \right) = v(t)$$

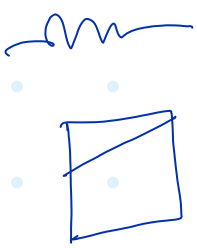
$$v(t) = R_L\, i(t) - e_h$$

$$R_L = \frac{L}{h} \qquad e_h = R_L\, i(t-h)$$

$$i(t) = \frac{v(t)}{R_L} + \frac{e_h}{R_L}$$

$$i(t) = \frac{v(t)}{R_L} + i_h$$

$$i_h = i(t-h) = \frac{e_h}{R_L}$$



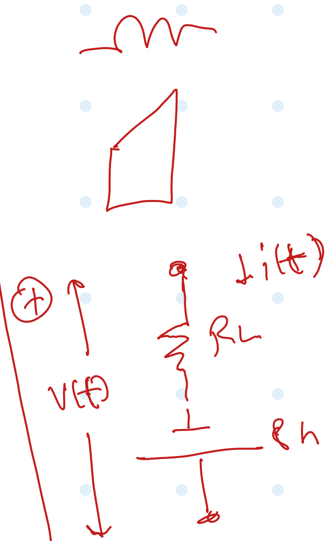
$$v(t) = R_L\, i(t) - e_h$$

$$i(t) = \frac{1}{R_L} v(t) + i_h$$

$$R_L = L/h$$

$$e_h = R_L\, i(t-h)$$

$$i_h = i(t-h) = e_h/R_L$$



$$v(t) = R_L i(t) - e_h$$

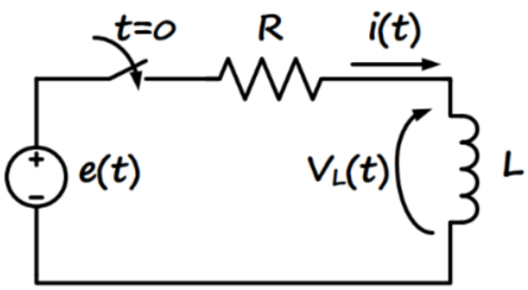
$$i(t) = \frac{v(t)}{R_L} + i_h$$

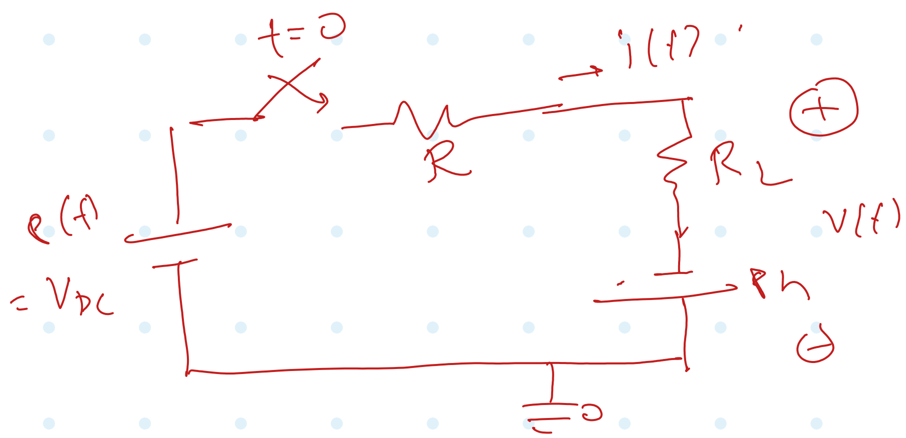
$$R_L = \frac{2L}{h}$$

$$e_h = R_L i(t-h) + v(t-h)$$

$$i_h = i(t-h) + \frac{v(t-h)}{R_L}$$

$$= e_h / R_L$$





$$V_{DC} - i(t)\cdot(R + R_L) + e_h = 0$$

$$\boxed{i(t) = \frac{V_{DC} + e_h}{R + R_L}}$$

$$v(t) = R_L i(t) - e_h$$

$$\frac{R_L}{R+R_L}(V_{DC} + e_h) - e_h = \frac{R_L}{R+R_L} V_{DC} - \frac{R}{R+R_L} e_h$$

$$v(0^-) = 0 \qquad i(0^-) = 0$$

$$= \frac{R_L}{R+R_L}V_{DC} - \frac{R}{R+R_L}\varphi_h$$

t	$v(t) = R_L i(t) - \varphi_h$	$i(t) = \frac{V_{DC}+\varphi_h}{R+R_L}$ $(t>0)$	$\varphi_h = v(t-h) + R_L i(t-h)$	$i_h = \frac{\varphi_h}{R_L}$
$-h$	0	0	—	—
0^+	$\left(\frac{R_L}{R+R_L}\right)V_{DC}$	$\left(\frac{1}{R+R_L}\right)V_{DC}$	0	0
h	$\left(\frac{R_L(R+3R_L)}{(R+R_L)^2} - \frac{2R_L}{R+R_L}\right)V_{DC}$	$\left(\frac{R+3R_L}{(R+R_L)^2}\right)V_{DC}$	$\left(\frac{2R_L}{R+R_L}\right)V_{DC}$	$\left(\frac{2}{R+R_L}\right)V_{DC}$

$$= \frac{R_L(R+3R_L) - 2R_L(R+R_L)}{(R+R_L)^2}\cdot V_{DC}$$

$$\frac{R_L}{R+R_L}V_{DC} - \frac{2R\cdot R_L V_{DC}}{(R+R_L)^2}$$

$$= \frac{-R_L R + R_L^2}{(R+R_L)^2}V_{DC}$$

$$\frac{R_L(R+R_L) - 2R R_L}{(R+R_L)^2}V_{DC}$$

$$= \frac{R_L(R_L-R)}{(R+R_L)^2}V_{DC}$$

h	$\frac{R_L(R_L-R)}{(R+R_L)^2}V_{DC}$		
h	$\frac{R_L(R_L-R)}{(R_L+R)}V_{DC}$	$\frac{3R_L+R}{(R_L+R)^2}V_{DC}$	$\left(\frac{2R_L}{R_L+R}\right)V_{DC}$

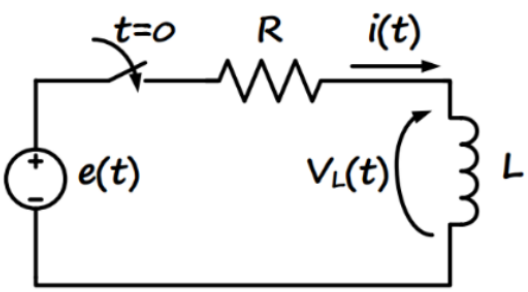
$$v(t) = R_L\, i(t) - e_h$$

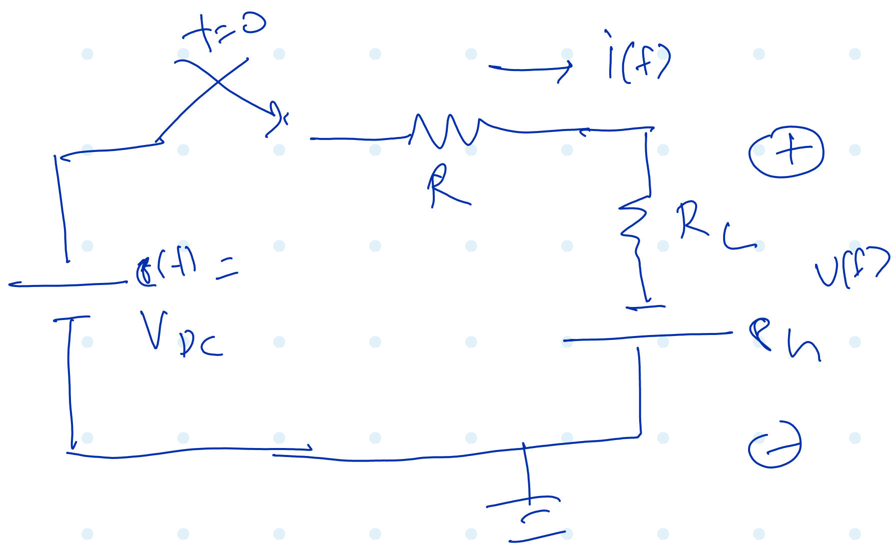
$$i(t) = \frac{1}{R_L} v(t) + i_h$$

$$R_L = L/h$$

$$e_h = R_L\, i(t-h)$$

$$i_h = i(t-h) = e_h/R_L$$





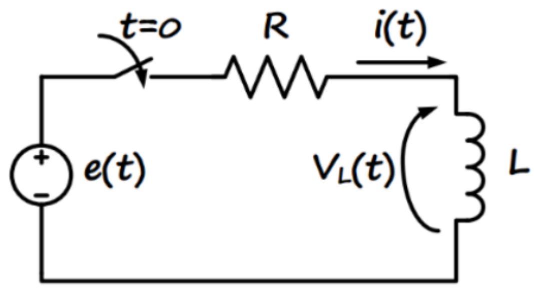
$$\boxed{i(t) = \frac{V_{DC} + e_h}{R + R_L}}$$

$$v(t) = V_{DC} - \frac{R}{R+R_L}\left(V_{DC} + e_h\right)$$

$$\text{or} \quad \boxed{v(t) = \frac{R_L}{R+R_L} V_{DC} - \frac{R}{R+R_L} e_h}$$

SECTION B (Basic Circuit Numerical Simulation)

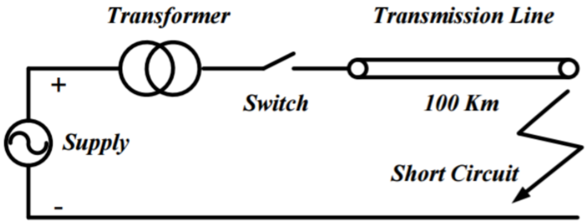
The parameters of the circuit below are: $R = 10\ \Omega$, $L = 20$ mH, $e(t) = 10$ V dc, and the switch is closed at t=0. The circuit below is to be solved from t=0 to t=10 ms using: a) PSCAD software, b) your own MATLAB code based on the nodal analysis method.

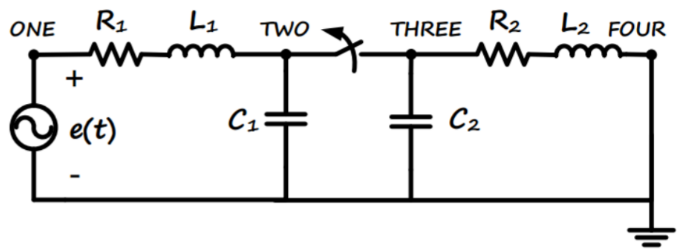


3) Show the results for $i(t)$ obtained in the six cases (PSCAD, your Backward Euler code, and your Trapezoidal code; each with two different step sizes) superimposed in a (big enough) Plot. Show the same for $v_L(t)$ in a separate plot. Comment on the accuracy of the methods, impact of the time-step size, and how the PSCAD results compare with your Trapezoidal code.

SECTION C (Circuit Interruption)

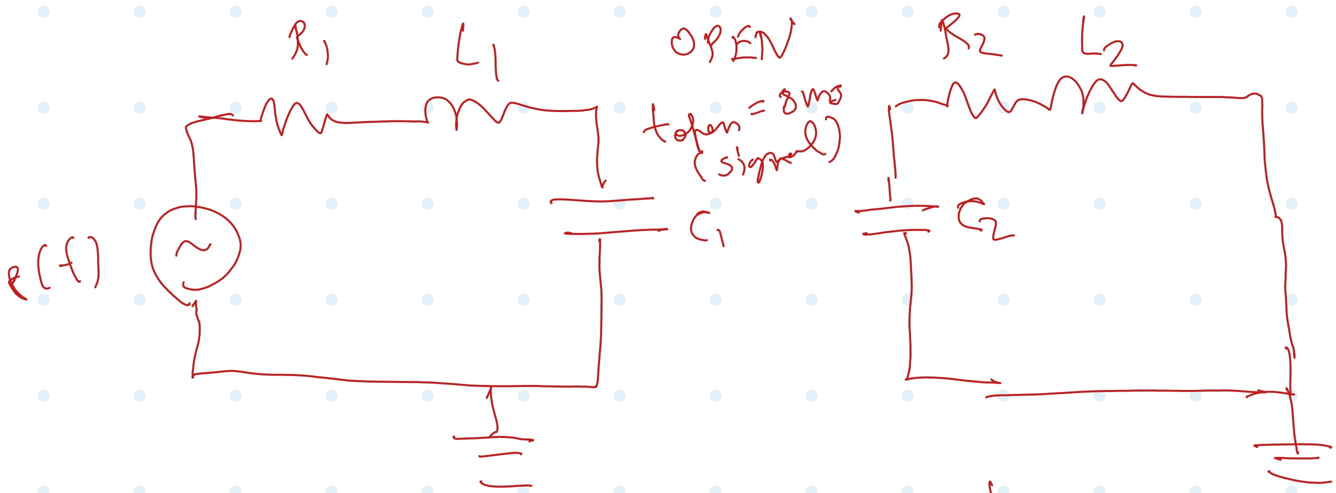
The case below (figure on the left) represents a typical short-circuit fault in transmission lines and the breaker operation for line interruption. The equivalent circuit is shown in the figure on the right



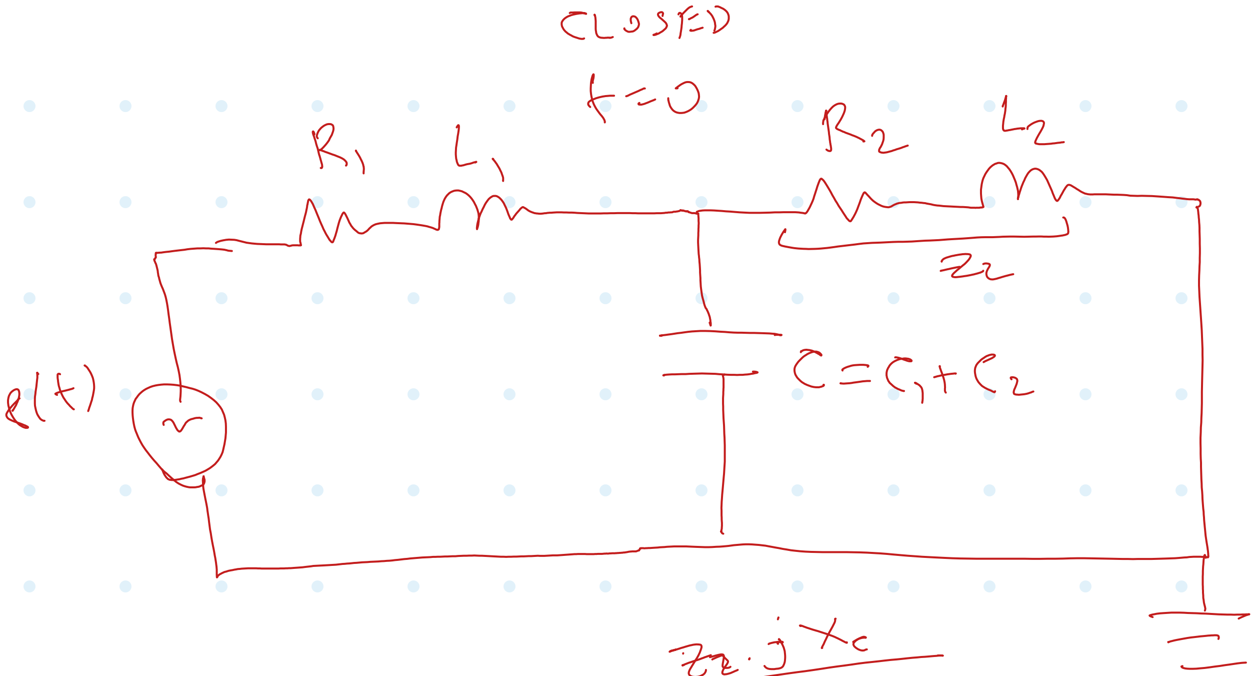


On the left-hand side of the switch/breaker, we have the source voltage and a resistance and inductance representing the supply system combined with the transformer. The capacitor to the left of the breaker represents the combined substation capacitance due to the substation busbars. On the right-hand side of the breaker, we have the transmission line parameters: resistance, inductance, and capacitance represented as simple lumped parameters, where the right side is short-circuited due to the fault, assuming that the fault has already been applied. We have assumed a single-phase circuit here for simplicity. The parameters of the circuit below are: $e(t)$ = 230×cos(ωt) kV, ω=377 rad/s, R_1=3 Ω, L_1=350 mH, R_2=50 Ω, L_2=100 mH, C_1=10 nF, C_2=600 nF. Assume that the system is operating with zero initial conditions and the breaker is initially closed. At t=8 ms, the breaker is commanded to open the line, but it will not actually open until the current passes through zero. The study period is from t=0 to t=20 ms.

1) Choose an appropriate simulation time-step size Δt (assuming the Trapezoidal integration rule) according to the time constants in the circuit. For this, evaluate the approximate resonant frequencies analytically before and after the breaker operation.



$\frac{1}{\sqrt{L_1 C_1}}$ $\frac{1}{\sqrt{L_2 C_2}}$

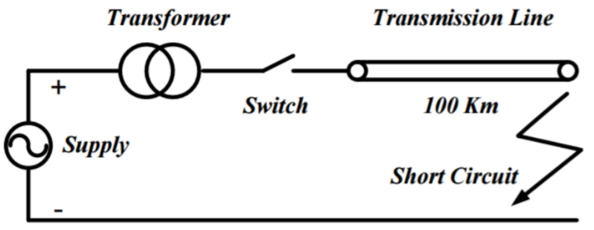


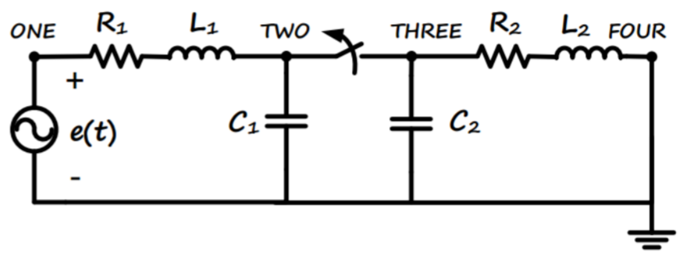
$$Z_{Th} = R_1 + jX_{L_1} + \frac{Z_2 \cdot jX_C}{Z_2 + jX_C}$$

$$Z_{Th} = R_1 + j\omega L_1 + \frac{(R_2 + j\omega L_2) \cdot \left(j\frac{1}{\omega C_2}\right)}{(R_2 + j\omega L_2) + j\left(\frac{1}{\omega C_2}\right)}$$

SECTION C (Circuit Interruption)

The case below (figure on the left) represents a typical short-circuit fault in transmission lines and the breaker operation for line interruption. The equivalent circuit is shown in the figure on the right





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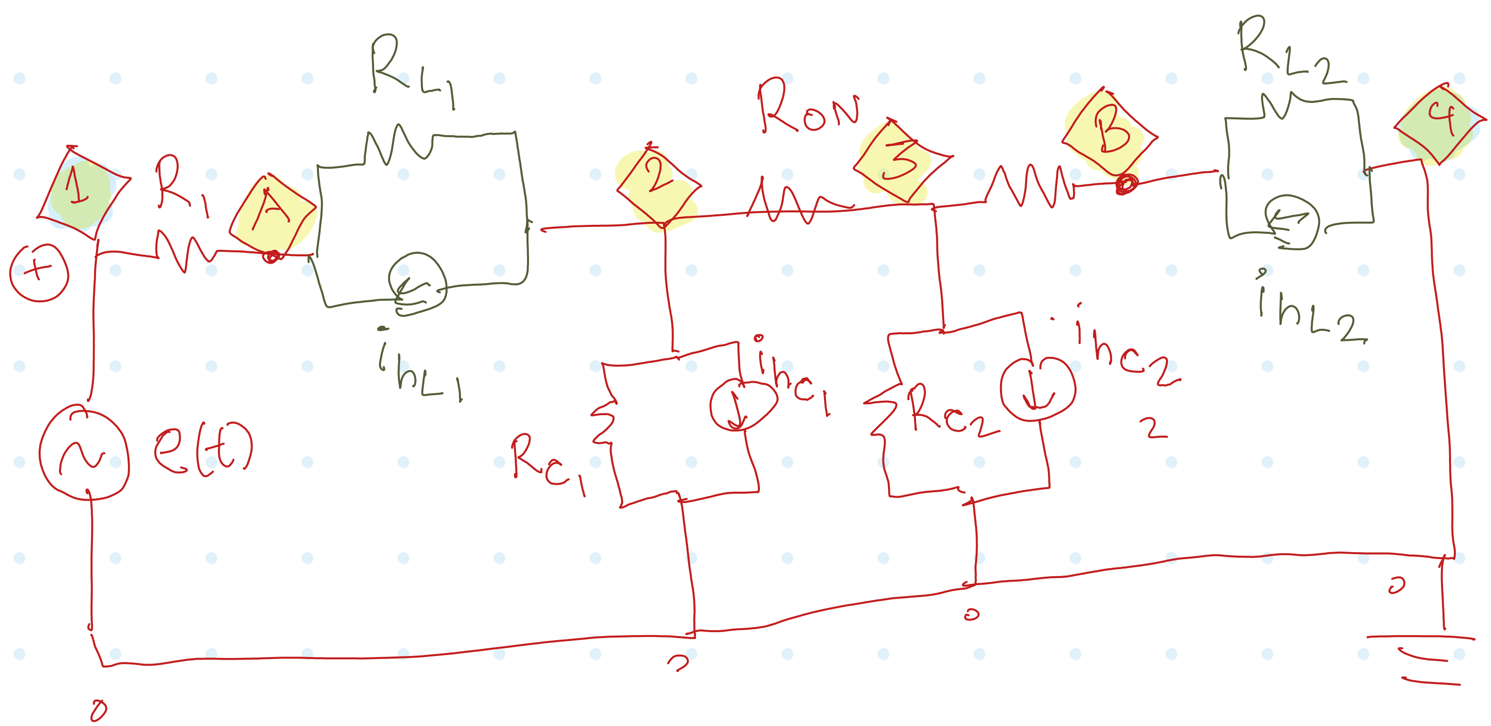
2) Write your own MATLAB code to solve the system and also obtain the solution using PSCAD.



G_{AA} $\quad$ G_{AB} $\quad$ $\vec{V}_A$

$$
\begin{array}{c}
\\ A \\ 2 \\ 3 \\ B \\ 1 \\ 4
\end{array}
\begin{array}{c}
\begin{array}{cccccc} A & 2 & 3 & B & 1 & 4 \end{array} \\
\left[\begin{array}{cccc|cc}
\frac{1}{R_{L_1}}+\frac{1}{R_1} & -1/R_{L_1} & 0 & 0 & -1/R_1 & 0 \\
-1/R_{L_1} & 1/R_{C_1}+\frac{1}{R_{L_1}}+\frac{1}{R_{ON}} & -1/R_{ON} & 0 & 0 & 0 \\
0 & -1/R_{ON} & 1/R_{C2}+\frac{1}{R_{ON}}+\frac{1}{R_2} & -1/R_2 & 0 & 0 \\
0 & 0 & -1/R_2 & \frac{1}{R_{L_2}} & 0 & -1/R_{L_2} \\
\hline
-1/R_1 & 0 & 0 & 0 & X & X \\
0 & 0 & 0 & -1/R_{L_2} & X & X
\end{array}\right]
\end{array}
\left[\begin{array}{c}
V_A \\ V_2 \\ V_3 \\ V_B \\ \hline e(t) \\ 0
\end{array}\right]
\begin{array}{c}
\\ \\ \\ \\ \\ \vec{V}_B
\end{array}
$$

$$
\begin{array}{c}
A \\ 2 \\ 3 \\ B \\ 1 \\ 4
\end{array}
\left[\begin{array}{c}
i_{hL_1} \\ -i_{hL_1}-i_{hC_1} \\ -i_{hC_2} \\ i_{hL_2} \\ \hline \\ \\
\end{array}\right]
$$

For ckt off (when $i_{23} < \varepsilon$)
Replace R_{ON} w/ R_{OFF}!

$$
\underset{4\times 4}{[G_{AA}]}\ \underset{4\times 1}{\vec{V}_A(t)} = \underset{4\times 1}{\vec{i}_{hA}(t)} - \underset{4\times 2}{[G_{AB}]}\ \underset{2\times 1}{\vec{V}_B(t)}
$$

A

$$i(t) = \frac{1}{R_L} v(t) + i_n(t)$$

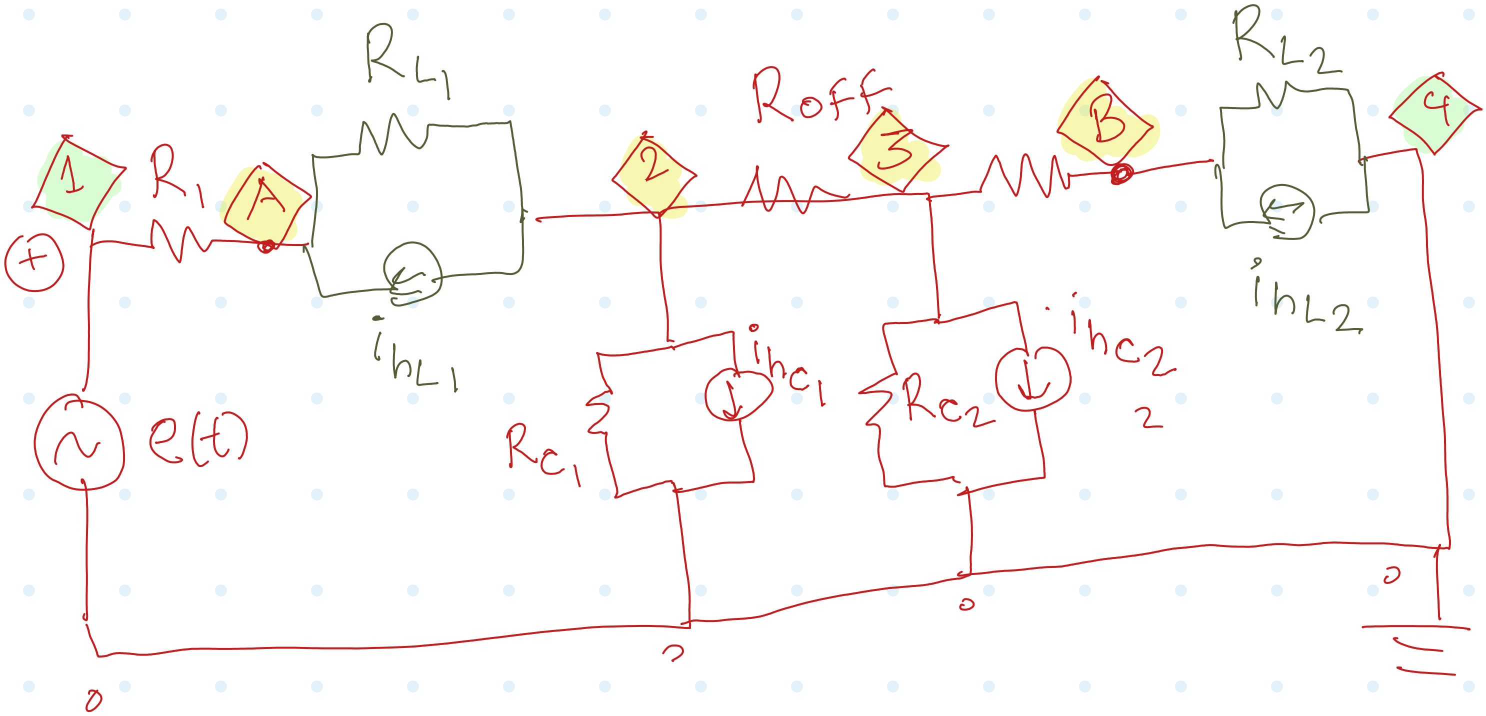
$$R_L = \frac{2L}{h} \qquad i_n(t) = \frac{1}{R_L} v(t-h) + i(t-h)$$

B

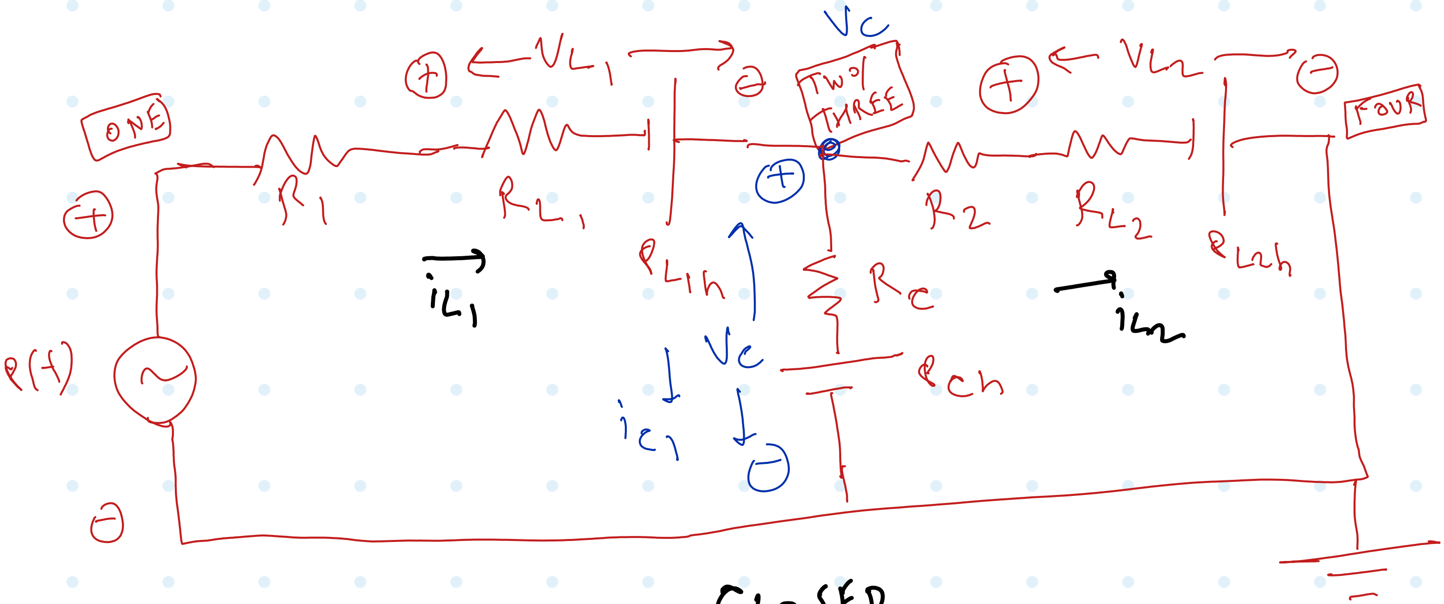
$$i(t) = \frac{1}{R_c} v(t) - i_h(t)$$

$$R_c = \frac{h}{2C} \qquad i_h(t) = \frac{1}{R_c} v(t-h) + i(t-h)$$

$$i_n(t) = \frac{e_n(t)}{R_c}$$

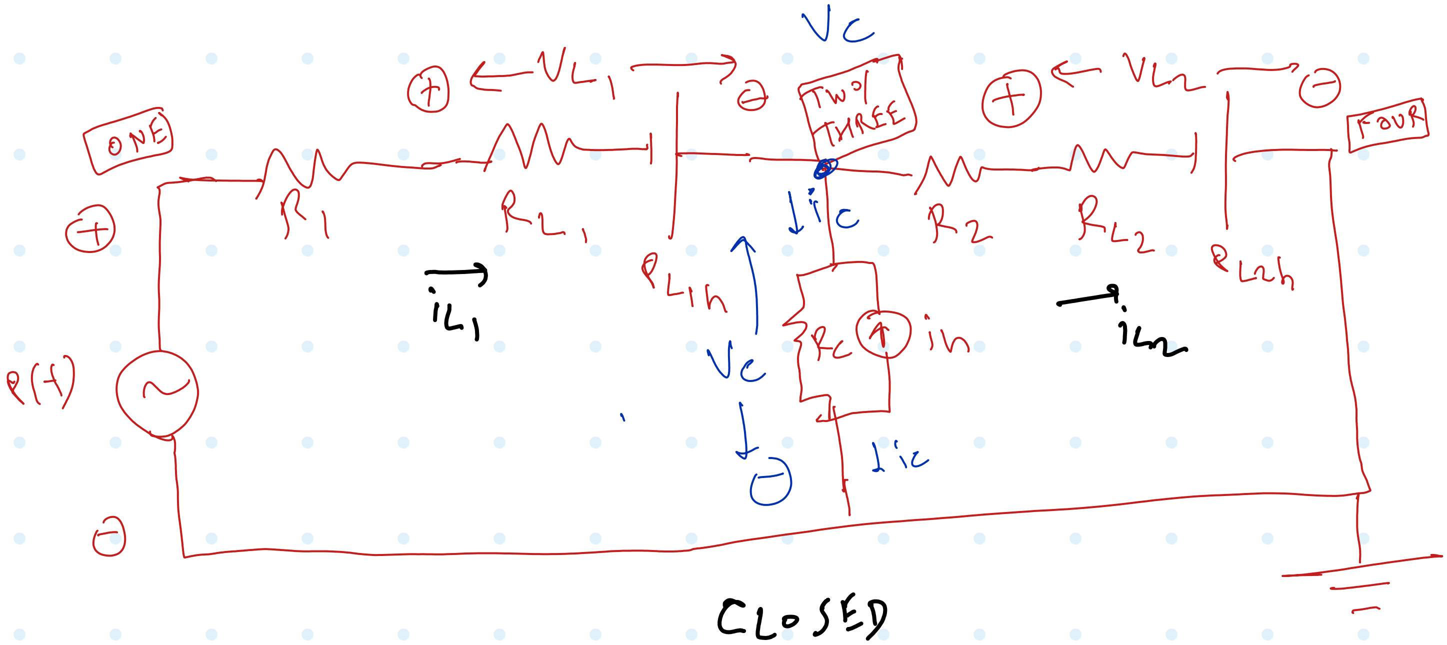


After switch off!

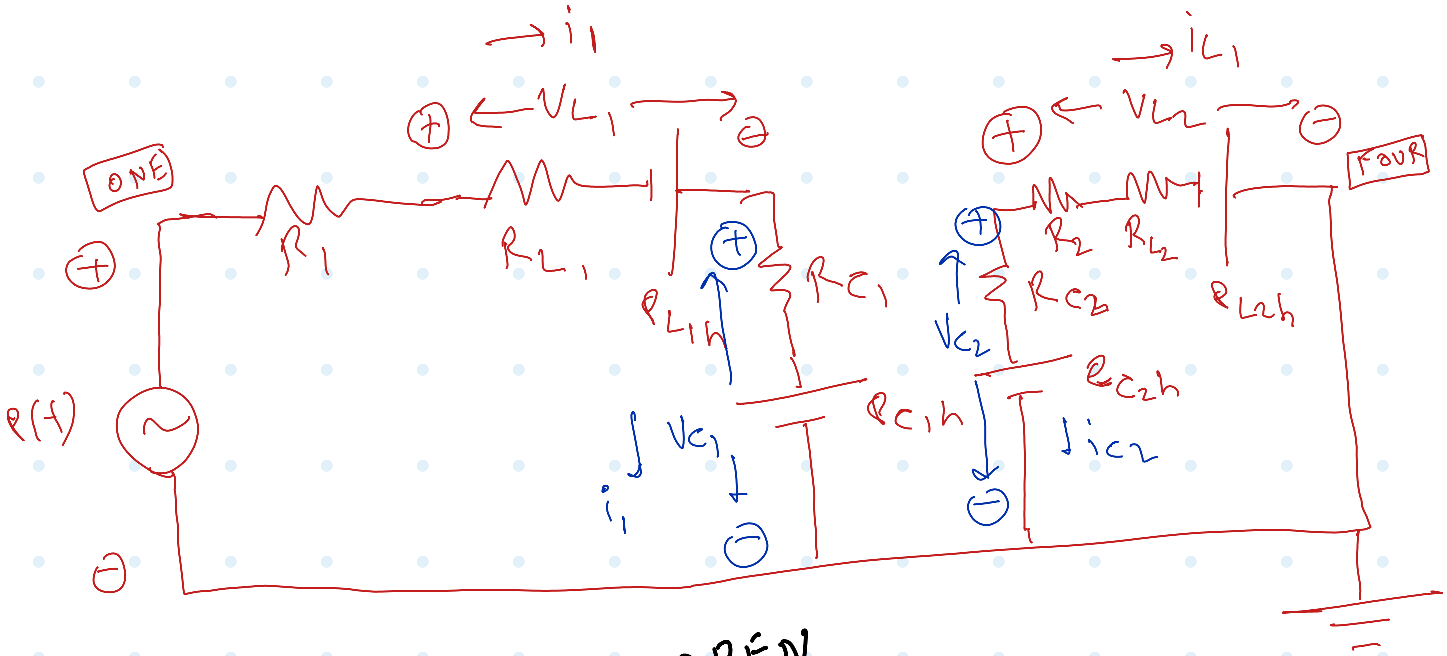


CLOSED

$t \geq 0$



$$e(t) - i_{L_1}(R_1 + R_{L_1}) + e_{L_1 h} = V_c$$



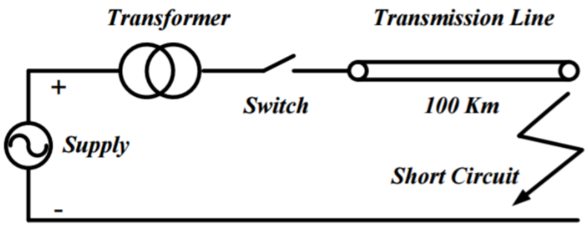
i_1
V_{L_1}
i_{L_1}
V_{L_2}
ONE
FOUR
R_1
R_{L_1}
R_{C_1}
R_2
R_{L_2}
R_{C_2}
R_{L_1h}
R_{L_2h}
R_{C_1h}
R_{C_2h}
V_{C_1}
V_{C_2}
i_1
i_{C_2}
$e(t)$

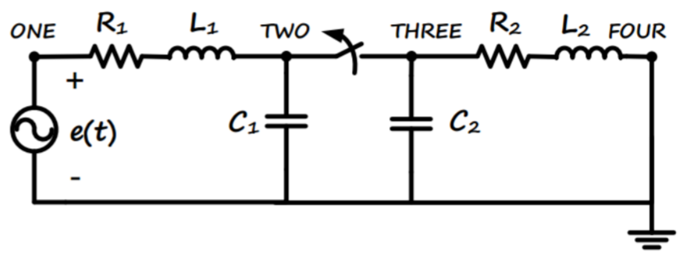
OPEN

$+78+\sim$

SECTION C (Circuit Interruption)

The case below (figure on the left) represents a typical short-circuit fault in transmission lines and the breaker operation for line interruption. The equivalent circuit is shown in the figure on the right





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3) Show the results for voltages at nodes *Two* and *Three* and the fault current (i.e., from node *Four* to *ground*) in three separate (big enough) plots. In each plot, superimpose the results from your code and also from PSCAD, both obtained using the same simulation time-step size obtained in part 1) for comparison. Comment on the accuracy of your results compared to those of PSCAD.